

Request for Proposal

By IIT Roorkee, powered by Futureense

Capstone Project for AI Ops Certification

Advanced PG Certification in AI Engineering on Cloud and AIOps

Generative AI + Agentic AI with Cloud Deployment

Submission Deadline: 25 June 2026

Project Duration: 4 Weeks

Team Size: 5–6 learners per team

Mode of Submission: LMS submission under Module 15

1. Disclaimer and Project Selection Guidelines

This document outlines the capstone project themes for the AI Ops Certification final project. Each team must select **any one project theme** from the list provided in this RFP document.

The groups have already been created and shared with learners. Each group must work collaboratively in teams of **5 to 6 members** and submit a complete capstone package by the final deadline.

Learners are expected to treat this capstone as a real-world AI engineering proposal and implementation project. The objective is not only to build an AI chatbot or agent locally, but to demonstrate a complete, working, cloud-deployed AI solution using agentic workflows, RAG pipelines, frontend integration, and AWS deployment.

The technical themes and expectations in this RFP are based on the provided capstone theme document for the AI Engineering on Cloud and AIOps program.

2. Capstone Objective

The objective of this capstone project is to help learners demonstrate their ability to design, build, and deploy an end-to-end AI solution using:

- Generative AI
- Agentic AI workflows
- Retrieval-Augmented Generation
- Multi-agent orchestration
- Cloud deployment on AWS
- Human-in-the-loop approval
- Practical frontend or dashboard integration

Each project should simulate a real-world enterprise use case where AI agents can assist teams in analyzing information, generating recommendations, automating workflows, summarizing documents, or supporting business decisions.

3. General Project Instructions

Each team must select **one theme** from the domains listed below:

1. Telecom
2. Healthcare
3. Finance

Teams are encouraged to create their own synthetic datasets, PDFs, documents, logs, reports, forms, manuals, or workflows depending on the chosen project theme.

Teams must ensure that their solution is feasible within the 4-week capstone timeline and can be demonstrated as a standalone working project.

4. Team and Collaboration Guidelines

Each team must:

- Work in the already assigned groups.
- Have 5–6 members per team.
- Divide responsibilities clearly across development, documentation, deployment, testing, and presentation.

- Maintain weekly progress documentation.
- Ensure that all members contribute meaningfully to the final submission.
- Track sprint-wise activities in the final report.

The same communication thread will be used for common project-related questions, announcements, clarifications, and updates.

5. Project Timeline

Week	Timeline	Expected Focus
Week 1	25 May – 31 May	Theme selection, problem understanding, dataset/document planning, architecture design
Week 2	1 June – 7 June	Data creation, RAG setup, agent workflow planning, basic prototype development
Week 3	8 June – 14 June	Multi-agent implementation, workflow integration, frontend/dashboard development
Week 4	15 June – 25 June	AWS deployment, testing, final report, PPT preparation, presentation recording

Final Submission Deadline: 25 June 2026

6. Mandatory Deliverables

Each team must submit the following:

A. Final Project Report

The report must document the complete project journey and must include:

1. Project title
2. Selected domain and theme

3. Problem statement
4. Proposed solution
5. Dataset/document/log creation approach
6. System architecture
7. Agent workflow design
8. RAG pipeline explanation
9. Tools and technologies used
10. AWS deployment details
11. Screenshots of application/dashboard/workflow
12. Testing and validation
13. Error handling and retry mechanisms
14. Human approval step
15. Sprint-wise progress documentation
16. Challenges faced and how they were resolved
17. Final conclusion
18. Individual contribution summary

The report must clearly mention what was completed in:

- Week 1
- Week 2
- Week 3
- Week 4

B. Final Presentation PPT

Each team must create a PPT covering:

1. Project title and team details
2. Problem statement
3. Selected theme and domain
4. Solution overview
5. Architecture diagram
6. Agent workflow
7. RAG pipeline
8. AWS deployment overview
9. Demo screenshots
10. Key outcomes
11. Challenges and learnings
12. Future scope

C. Recorded Presentation

Each team must record a presentation explaining the project. The recording should include:

- Problem explanation
- Solution walkthrough
- Architecture explanation
- Application/demo walkthrough
- Deployment explanation
- Key learnings

D. Project Files

Teams must submit all relevant files, including:

- Source code
 - Dataset files
 - Synthetic documents/PDFs/logs/forms used
 - Final report
 - Final PPT
 - Recorded presentation file or link
 - Deployment details or hosted application link, if available
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7. Common Technical Expectations Across All Projects

Each team should demonstrate the following capabilities:

- Multi-agent workflows
 - Sequential and parallel execution
 - Router-based decision making
 - RAG pipelines
 - Human approval step
 - Tool-using agents
 - Memory/context handling
 - Error handling and retries
 - Deployment on AWS
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8. Recommended Simplified Tech Stack

Layer	Suggested Tools
Agent Framework	LangGraph / CrewAI
LLM	OpenAI / Gemini
RAG	LangChain
Vector Database	ChromaDB / FAISS
Frontend	Streamlit
Deployment	AWS EC2
Storage	AWS S3
Containerization	Docker, optional

All capstone projects are expected to go beyond building local AI agents or chatbots. Teams must deploy their applications on AWS, primarily using **EC2 for hosting** and **S3 for document/file storage**. Docker-based containerization is optional.

9. Capstone Project Themes

Telecom Domain Projects

Theme 1: AI-Based Telecom Incident Ticket Analyzer

Problem Statement

Telecom teams receive many issue tickets daily and struggle to prioritize them.

Learners Can Create

- Synthetic ticket dataset
- Synthetic outage logs
- Sample telecom PDFs

AI Capabilities

- Ticket summarization
- Priority classification
- Suggested resolution generation

Agent Workflow

- Ticket Classification Agent
- Severity Detection Agent
- Resolution Suggestion Agent
- Human Approval Agent

Workflow Type

Sequential + Router workflow

RAG Opportunity

Use telecom troubleshooting PDFs created by learners.

AWS Deployment

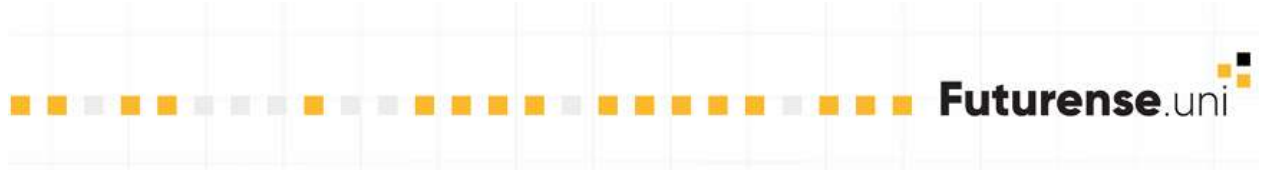
- Streamlit app deployed on AWS EC2
- PDFs stored in S3

Theme 2: Telecom Customer Complaint Resolution Assistant

Problem Statement

Customer complaints are difficult to analyze manually.

Learners Can Create



- Synthetic complaint chats
- Complaint history dataset

AI Features

- Complaint summarization
- Sentiment analysis
- Suggested response generation

Agent Workflow

- Sentiment Agent
- Complaint Categorization Agent
- Response Generation Agent

Workflow Type

Parallel + Sequential

RAG Opportunity

Complaint policy documents.

AWS Deployment

Deploy chatbot application on AWS EC2.

Theme 3: Telecom Network Outage RCA Assistant

Problem Statement

Engineers spend time identifying root causes during outages.

Learners Can Create

- Synthetic network logs
- Sample outage reports

AI Features

- Root-cause explanation

- Incident summary
- Suggested troubleshooting steps

Agents

- Log Analysis Agent
- RCA Agent
- Recommendation Agent

Workflow

- Parallel log analysis
- Router for escalation

AWS Deployment

Deploy dashboard/workflow on AWS EC2.

Theme 4: Telecom Fraud Alert Investigation Assistant

Problem Statement

Fraud alerts require manual investigation.

Learners Can Create

- Fake call records
- Synthetic fraud cases

AI Features

- Fraud summarization
- Risk scoring
- Investigation report generation

Agents

- Fraud Detection Agent
- Risk Analysis Agent
- Report Generation Agent

Workflow

Sequential workflow with human checkpoint.

AWS Deployment

Deploy investigation assistant on AWS.

Theme 5: AI-Powered Telecom Knowledge Assistant

Problem Statement

Engineers struggle to search for long telecom manuals.

Learners Can Create

- Telecom SOP PDFs
- Device manuals

AI Features

- RAG chatbot
- Context-aware answers
- Document summarization

Agents

- Retrieval Agent
- Answer Validation Agent
- Summarization Agent

Workflow

RAG + memory-aware chatbot.

AWS Deployment

Deploy chatbot on AWS EC2.

Healthcare Domain Projects

Theme 6: AI Clinical Report Summarization Assistant

Problem Statement

Doctors spend time reading long reports.

Learners Can Create

- Synthetic patient reports
- Lab test PDFs

AI Features

- Report summarization
- Highlight abnormal findings
- Follow-up recommendations

Agents

- Report Analysis Agent
- Summary Agent
- Recommendation Agent

Workflow

Sequential workflow.

RAG

Medical guideline PDFs.

AWS Deployment

Deploy Streamlit application on AWS EC2.

Theme 7: AI Medical Research Paper Assistant

Problem Statement

Researchers struggle to analyze long papers.

Learners Can Create

- Synthetic research PDFs
- Medical article dataset

AI Features

- Research summarization
- Key findings extraction
- Q&A chatbot

Agents

- Paper Reader Agent
- Insight Agent
- Citation Agent

Workflow

Parallel extraction workflow.

AWS Deployment

Deploy research assistant on AWS EC2.

Theme 8: Hospital Resource Allocation Assistant

Problem Statement

Hospitals struggle to allocate beds and staff.

Learners Can Create

- Synthetic hospital datasets

AI Features

- Resource allocation suggestions
- Emergency prioritization

Agents

- Bed Allocation Agent
- Staff Allocation Agent
- Emergency Routing Agent

Workflow

Router-based workflow.

AWS Deployment

Deploy dashboard on AWS.

Theme 9: AI Patient Follow-Up Assistant

Problem Statement

Hospitals miss follow-up communication.

Learners Can Create

- Synthetic patient follow-up data

AI Features

- Personalized reminders
- Follow-up summaries

Agents

- Patient Summary Agent
- Reminder Agent
- Escalation Agent

Workflow

Sequential workflow with memory.

AWS Deployment

Deploy follow-up system on AWS EC2.

Theme 10: Healthcare Insurance Claim Review Assistant

Problem Statement

Insurance claims are manually reviewed.

Learners Can Create

- Synthetic claims PDFs
- Fake insurance forms

AI Features

- Claim summarization
- Missing field detection
- Approval recommendation

Agents

- OCR Agent
- Validation Agent
- Recommendation Agent

Workflow

Sequential + human approval.

AWS Deployment

Deploy review assistant on AWS.

Finance Domain Projects

Theme 11: AI Loan Eligibility Assessment Assistant

Problem Statement

Banks manually review loan applications.

Learners Can Create

- Synthetic customer financial data
- Loan application PDFs

AI Features

- Loan summarization
- Risk explanation
- Eligibility recommendation

Agents

- Financial Analysis Agent
- Risk Agent
- Recommendation Agent

Workflow

Sequential + router workflow.

AWS Deployment

Deploy assessment system on AWS EC2.

Theme 12: AI Financial Document Q&A Assistant

Problem Statement

Users struggle to understand financial reports.

Learners Can Create

- Annual reports
- Fake balance sheets

AI Features

- RAG chatbot



- Financial summarization
- KPI extraction

Agents

- Retrieval Agent
- KPI Extraction Agent
- Summary Agent

Workflow

RAG workflow with memory.

AWS Deployment

Deploy chatbot on AWS EC2.

Theme 13: AI Fraud Transaction Investigation Assistant

Problem Statement

Fraud analysts manually inspect suspicious transactions.

Learners Can Create

- Synthetic transaction dataset

AI Features

- Fraud explanation
- Risk scoring
- Investigation reports

Agents

- Transaction Agent
- Fraud Detection Agent
- Reporting Agent

Workflow

Parallel fraud analysis.

AWS Deployment

Deploy fraud analysis dashboard on AWS.

Theme 14: AI Expense Audit Assistant

Problem Statement

Expense auditing takes significant manual effort.

Learners Can Create

- Synthetic invoices
- Fake reimbursement forms

AI Features

- Expense summarization
- Duplicate detection
- Policy violation detection

Agents

- OCR Agent
- Validation Agent
- Audit Agent

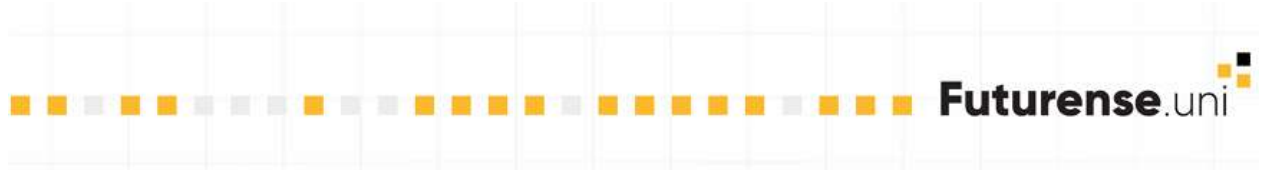
Workflow

Sequential workflow.

AWS Deployment

Deploy audit assistant on AWS EC2.

Theme 15: AI Investment Research Assistant



Problem Statement

Analysts spend time reviewing multiple market reports.

Learners Can Create

- Synthetic market reports
- Financial news PDFs

AI Features

- Market trend summarization
- Investment insights
- Comparative analysis

Agents

- Market Research Agent
- Trend Analysis Agent
- Summary Agent

Workflow

Parallel research workflow.

AWS Deployment

Deploy research assistant on AWS.

10. AWS Deployment Expectations

Each team should deploy their project using basic AWS services.

Component	Suggested AWS Service
Application Hosting	EC2

File Storage	S3
Containerization	Docker, optional

The final project should demonstrate a working end-to-end cloud-deployed AI solution. Teams should be able to explain how the application is hosted, where the files/documents are stored, and how the AI workflow is triggered through the frontend or dashboard.

11. Suggested Report Structure

Teams may follow the structure below for their final report:

1. Cover Page

- Project title
- Domain
- Theme selected
- Team name
- Names of all team members
- Submission date

2. Executive Summary

Briefly explain the problem, solution, and final outcome.

3. Problem Statement

Explain the selected business problem in detail.

4. Project Objectives

Mention the goals of the project.

5. Dataset and Document Creation

Explain what synthetic datasets, PDFs, logs, forms, or reports were created.

6. System Architecture

Include architecture diagram and explain all components.

7. Agent Workflow Design

Explain all agents used in the project and how they interact.

8. RAG Pipeline

Explain document ingestion, embedding, vector storage, retrieval, and response generation.

9. Human Approval Step

Explain where human validation or approval is included.

10. Memory and Context Handling

Explain how conversation history, user context, or workflow memory is managed.

11. Error Handling and Retry Mechanism

Explain how the system handles failures or incorrect inputs.

12. AWS Deployment

Explain EC2 deployment, S3 storage, and any optional Docker usage.

13. Sprint-Wise Progress

Week 1 Progress

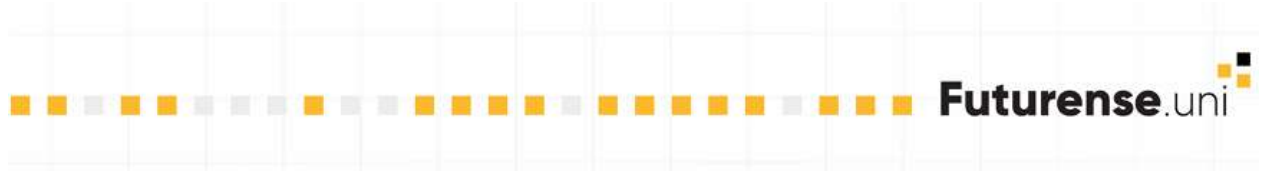
Mention planning, theme selection, dataset preparation, and architecture design.

Week 2 Progress

Mention implementation of data pipeline, RAG setup, and initial agent development.

Week 3 Progress

Mention multi-agent integration, frontend/dashboard creation, and workflow testing.



Week 4 Progress

Mention AWS deployment, final testing, report completion, PPT preparation, and presentation recording.

14. Screenshots and Demo Evidence

Add screenshots of the working application, dashboard, chatbot, AWS deployment, and outputs.

15. Challenges Faced

Mention technical, data, deployment, or workflow-related challenges.

16. Key Learnings

Mention what the team learned from the capstone.

17. Future Scope

Mention how the project can be improved further.

18. Individual Contributions

Mention each team member's contribution clearly.

12. Suggested PPT Structure

Teams may follow the structure below:

1. Title Slide
2. Team Members
3. Selected Domain and Theme
4. Problem Statement
5. Proposed Solution
6. Dataset/Documents Used
7. Architecture Diagram
8. Agent Workflow
9. RAG Pipeline

10. Human Approval Flow
11. AWS Deployment
12. Application Screenshots
13. Demo Walkthrough
14. Challenges Faced
15. Key Learnings
16. Future Scope
17. Thank You

13. Evaluation Expectations

Projects will be reviewed based on:

Evaluation Area	Expected Evidence
Problem Understanding	Clear explanation of the selected theme and business problem
AI Workflow Design	Proper use of agents, workflows, routing, and orchestration
RAG Implementation	Use of documents/PDFs/logs with retrieval-based answers
Technical Implementation	Working application, clean logic, and meaningful outputs
AWS Deployment	EC2/S3 usage and cloud-hosted demonstration
Documentation	Complete final report with sprint-wise progress
Presentation	Clear PPT and recorded explanation
Team Contribution	Balanced and documented contribution from team members

14. Evaluation Rubric

The capstone project will be evaluated based on the following rubric:

Evaluation Criteria	Weightage	Description
Problem Statement and Scope	10%	Clarity of the selected problem statement, relevance of the use case, and whether the project scope was realistic and achievable within the given 4-week timeline.
Targets Defined	20%	How clearly the team defined project goals, expected outputs, success criteria, technical milestones, and sprint-wise targets at the beginning of the project.
Targets Achieved	20%	Extent to which the team achieved the planned targets, completed the proposed features, and demonstrated measurable progress against the original project goals.
Code Implementation	/ 30%	Quality of technical implementation, agent workflow design, RAG pipeline, AWS deployment, frontend/dashboard integration, error handling, and overall working functionality.
Presentation Quality	20%	Clarity, structure, professionalism, demo explanation, PPT quality, recorded presentation quality, and ability to communicate the project effectively.
Total: 100%		

15. LMS Submission Guidelines

A submission provision will be created on the LMS under **Module 15** for the capstone project submission.

The following documents and details will be made available under **Module 15**:

1. Capstone RFP Document
2. Question Paper / Capstone Instructions
3. Supporting project submission file/instructions
4. Submission deadline and related details

The submission process will be similar to the regular assignment submission process on the LMS.

Each team must submit their final capstone deliverables through the submission provision created under **Module 15**. This provision will be created eventually, and learners will be informed once it is available.

Further submission guidelines and reminders will also be shared again closer to the deadline.

15.1 Submission Format

Each team can submit the deliverables in one of the following ways:

Option 1: ZIP File Submission

Teams may upload a single ZIP file containing all final deliverables.

The ZIP file should include:

- Final project report
- Final PPT
- Recorded presentation file or recording link
- Source code
- Dataset files
- Synthetic documents / PDFs / logs / forms used
- AWS deployment screenshots or proof
- Any additional supporting files

Option 2: Google Drive Folder Link

Teams may also create a Google Drive folder and submit the folder URL on the LMS.

If submitting a Google Drive folder link, teams must ensure that:

- All final deliverables are uploaded inside one properly organized folder.
- The folder access permission is correctly enabled.
- The sharing setting should allow access to anyone with the link.
- The folder should remain accessible until the evaluation process is completed.
- The Google Drive folder link should be submitted on the LMS under Module 15.

Recommended folder structure:

None

Team_Name_Capstone_Submission/

|

|— Final_Report.pdf

|— Final_Presentation.pptx

|— Recorded_Presentation_Link.txt

|— Source_Code/

|— Dataset/

|— Synthetic_Documents/

|— AWS_Deployment_Proof/

|— Additional_Files/

15.2 Team-Wise Submission Rule

Only **one person per team** needs to submit the capstone project on the LMS.

All team members do **not** need to submit individually.

The team should mutually decide who will submit the final project on behalf of the group.

15.3 Mandatory First Page Instruction

The **first page of the final report** must clearly mention the names of all team members.

The first page should include:

- Project title
- Selected domain
- Selected theme
- Team name, if applicable
- Names of all team members
- Email IDs / enrollment IDs, if required
- Submission date

This is important because the submission will be made by only one person per team, but the evaluation needs to correctly map all members of the group.

15.4 LMS Submission Note

The submission provision will be created under **Module 15** on the LMS.

Once the provision is created, learners will be able to submit their ZIP file or Google Drive folder link there, just like a regular assignment submission.

The final submission must be made **only** through the LMS provision created under Module 15.

16. Final Submission Checklist

Before submitting, each team must ensure the following are ready:

Deliverable	Required
Final project report	Yes
Final PPT	Yes
Recorded presentation	Yes
Source code	Yes
Dataset / synthetic files used	Yes
Synthetic documents / PDFs / logs / forms	Yes, if used
AWS deployment proof or screenshots	Yes
Google Drive link or ZIP file	Yes
Team member names on first page	Yes
Sprint-wise progress documentation	Yes
Individual contribution summary	Yes

17. Important Notes for Learners

- Only one theme must be selected by each team.
- Teams must not submit only a theoretical report.
- The project must include a working implementation.
- The application must be deployed on AWS.
- Synthetic data is allowed and encouraged.
- Teams may create their own PDFs, logs, reports, forms, documents, or datasets.
- The final report must include sprint-wise documentation.
- The final PPT and recorded presentation are mandatory.

- Only one person per team needs to submit the project on the LMS.
 - The first page of the report must include the names of all team members.
 - A submission provision will be created under Module 15 on the LMS.
 - Further submission guidelines and reminders will be shared again closer to the deadline.
 - The same communication thread will be used for common questions and announcements.
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18. Closing Note

This capstone is designed to help learners apply AI engineering, AIOps, cloud deployment, RAG, and agentic AI concepts in a practical enterprise-style project. Teams should focus on building a solution that is clear, usable, technically sound, and demonstrable.

The final outcome should show how Generative AI and Agentic AI can be applied to solve real-world operational problems across Telecom, Healthcare, or Finance domains.

The final deadline for submission is **25 June 2026**.